

6 (a) ATP is needed for many metabolic processes in living organisms.

(i) Describe the properties of ATP that make it suitable for its role as the universal energy currency.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) Suggest why ATP is needed for protein synthesis.

.....

.....

.....

.....

..... [2]

(b) Fig. 6.1 is a diagram of mitochondrion.

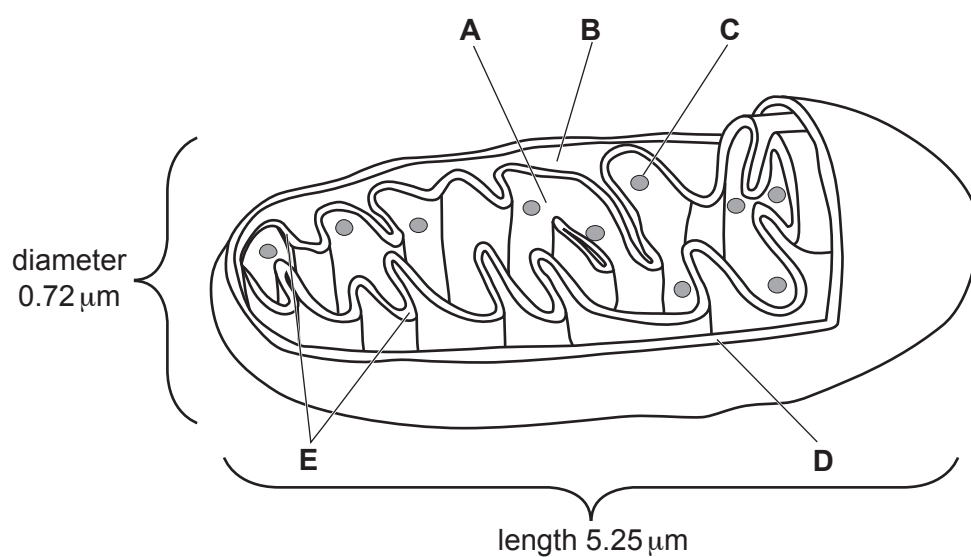


Fig. 6.1

- (i) Complete Table 6.1 using the letters **A** to **E** from Fig. 6.1.

Each letter may be used once, more than once, or not at all.

Table 6.1

statement	letter
the site of the Krebs Cycle	
a phospholipid bilayer impermeable to H ⁺ ions	
the site of translation	

[2]

- (ii) Assume that the mitochondrion in Fig. 6.1 is a cylinder.

Calculate the surface area of this mitochondrion.

Use the formula: Surface area of cylinder = $2\pi r^2 + 2\pi rh$

Show your working.

surface area = μm^2 [2]

- (iii) The inner membrane of the mitochondrion has a much larger surface area than the outer membrane because of the presence of cristae.

Different cell types vary in the number of cristae present per mitochondrion. Cardiac muscle cells have mitochondria with a very large number of cristae.

Suggest **and** explain why cardiac muscle cells have mitochondria with very large numbers of cristae.

.....

.....

.....

.....

..... [2]

[Total: 12]